SQL to KQL Hands-On Lab - Microsoft Fabric

This hands-on lab workbook is designed to help learners transition from SQL to KQL (Kusto Query Language) using practical exercises in Microsoft Fabric. Each lab includes a SQL example, the equivalent KQL query, and optional visualization tasks using KQL Querysets.

Lab 1: Basic Column Selection

SQL:

```
SELECT Name, Age FROM Employees;
```

KQL:

```
Employees
| project Name, Age
```

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 2: Filtering Data

SQL:

```
SELECT * FROM Employees WHERE Department = 'HR';
```

KQL:

```
Employees
| where Department == "HR"
```

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 3: Ordering Results

SQL:

```
SELECT * FROM Employees ORDER BY Age DESC;
```

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KQL:
Employees
| sort by Age desc

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 4: Top N Rows

SQL:
SELECT TOP 5 * FROM Sales;

KQL:
Sales
| top 5 by SalesAmount desc

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 5: Grouping & Aggregation

SQL:
SELECT Department, COUNT(*) FROM Employees GROUP BY Department;

KQL:
Employees
| summarize TotalEmployees = count() by Department

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

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Lab 6: Having Clause

SQL:
SELECT Department, COUNT(*) FROM Employees GROUP BY Department HAVING COUNT(*) > 10;

KQL:
```
Employees
| summarize Total = count() by Department
| where Total > 10
```

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 7: Join Tables

SQL:
SELECT * FROM Orders O JOIN Customers C ON O.CustomerID = C.CustomerID;

KQL:
```
Orders
| join kind=inner (
   Customers
) on CustomerID
```

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 8: Date Filtering

SQL:
SELECT * FROM Sales WHERE OrderDate >= '2024-01-01';

KQL:

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```
Sales
| where OrderDate >= datetime(2024-01-01)
```

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 9: Conditional Logic (CASE WHEN)

SQL:

```
SELECT Name, CASE WHEN Age >= 18 THEN 'Adult' ELSE 'Minor' END as Category FROM Employees;
```

KQL:

```
Employees
| project Name, Category = iif(Age >= 18, "Adult", "Minor")
```

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

Lab 10: Distinct Values

SQL:

```
SELECT DISTINCT Department FROM Employees;
```

KQL:

```
Employees
| distinct Department
```

🔍 Task: Run the above KQL query in a KQL Queryset in Microsoft Fabric.

📊 Optional: Add `| render <chart>` to visualize the results.

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Visualization Examples

Column Chart

```
Sales
| summarize Total = sum(Amount) by Region
| render columnchart
```

Pie Chart

```
Sales
| summarize Total = sum(Amount) by Product
| render piechart
```

Time Series (Line Chart)

```
Sales
| summarize TotalSales = sum(Amount) by bin(OrderDate, 1d)
| render timechart
```

Table View

```
Products
| project ProductID, Name, Price
| render table
```

Scatter Chart

```
Products
| project Price, Rating
| render scatterchart
```

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